Seminar on

Scope & Dimension of Agar (*Aquilaria spp.*) Plantation in N.E. Region

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ABSTRACTS OF PAPERS

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STUDIES ON RHIZOSPHERE MYCOFLORA OF AGAR PLANT IN ASSAM

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ABSTRACT

Agar plant or Sanchi plant (*Aquilaria spp.*) of Thymelaeceae family is becoming popular as naturally grown and as commercially cultivated plant in different districts of Assam. Agarwood or agaru and agar oil are extracted from the infected wood of the Agar tree. Agaru and agar oil are mostly used in perfume industry in various cosmetic preparation and as herbal medicine as utilized in Ayurvedic and Unani medicines. As observed in farmers field, infection starts from four years and harvest may be made from 12-15 years and may be extended upto 40 years under natural conditions. Mycrobial studies in the rhizosphere region of the agar plant upto one year shows the total fungal count from 6.27-6.87 $X10^4$ C. Fu/gm bacterial count 6.37 6.93 X 10^6 C Fu/gm and total actinomycetes count ranging from 3.05 3.53 X 10^5 C. Fu/gm of un-ammended soil. However, total microbial population of each group increased after incorporation of organic ammendments resulting in increase of *Trichoderma spp, Bacillus spp., Pseudomonas spp* and *Streptomyces spp* and population of plant growth promoting rhizobacteria (P G P R). These antagonists and PGPR can be further utilized as biopesticides and microbial inoculant for proper growth and development in establishing commercial cultivation of Agar tree.

FREE AMINO ACIDS DISTRIBUTION IN LEAF AND WOOD EXTRACTS OF *AQUILARIA MALACCENSIS* Lamk.

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ABSTRACT

The distribution pattern of free amino acids in leaf extracts of three variants and diseased and non-diseased wood extract of *Aquilaria malaccensis* lamk. were studied from taxonomic and physiological point of view. A total of 14 amino acids were detected in the leaf extracts of *A. malaccensis* variants. Of which the variant I was different from the rest by the presence of Glutamic acid, the variant II by the presence of Leucine, while the variant III was distinct from the variant I by the presence of Iso-leocine, Methionine and an unidentified amino acid corresponding to the spot No. 13 and from the variant II by the presence of Glycine and Theonine. However, the Variant I was more closely related to the Variant III (54.54 %) than that of the Variant II (33.33 %). Likewise, a total of 11 amino acids were detected in the wood extract of the species, of which 9 amino acids were present in the non-diseased and 7 amino acids in the diseased woods, while 5 amino acids were found to be of common occurrence for both diseased and non-diseased woods.

GERMINATION BEHAVIOUR AND STORAGE OF SEED AND SEEDLING VIGOUR OF *Aquilaria agallocha* ROXB.

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ABSTRACT

Germination behaviour, growth and seedling vigour of *Aquilaria agallocha* Roxb. were investigated under Hojai condition of Assam. Germination of agar seed is epigeal. Matured seeds with higher test weight exhibited higher percentage of germination in sand media (90 %). Fresh seeds obtained by splitting open the capsule showed a short dormancy period ranging from one week to four weeks. Maximum germination takes places within 20 days incase of fresh seeds and 7 days for stored seeds. Seed viability decreased with the increase of storage period and found to lose completely after 30 days of storage. Cold storage (4°C ±1°C) however, retained viability upto a period of 2 months or even more.

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A STUDY ON BIO-ECOLOGY OF *ZEUZERA CONFERTA* WALKER ON AGARWOOD PLANTATIONS

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ABSTRACT

Bio-ecology of *Zeuzera conferta* Walker has been studied in field condition of Regional Research Laboratory, Jorhat and surrounding agarwood growing areas in Jorhat district during 1997-1999. There was one generation a year and eggs, larval, pupal and adult stages lasted 16+3.65, 163+16.93, 23+2.10 and 13 +1.50 days respectively. In Jorhat region, overall 14.25 % of the trees were found to be infested. Average infestation of trees in Pukamura, Teok, Titabor, and Bhatemara area were 17.10 %, 12.10 %, 16.30 % and 11.50 % respectively. The borer infestation was maximum (60.70 %) in the plants of age group 6 -10 years. Infestation was in main trunk, but smaller branches were also tunneled occasionally. Infestation of borer was found to be higher (60.10 %) in monocultural plantation than that of mixtures (39.90 %).

RHIZOSPHERE MYCOFLORA OF *AQUILARIA AGALLOCHA* ROXB.

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ABSTRACT

Total 15 and 9 Nos. of fungi species have been isolated from the rhizosphere soil of *Aquilaria agallocha* Roxb. grown naturally in undisturbed area (sample 'B') and cultivated in disturbed area (sample 'A') respectively. *Mucor hiemalis, M. spinosus, Aspergillus ochraceous, Trichoderma album, T. koningii* are found in both sample 'A' and sample 'B'. In all dilutions (10-3, 10-5,10-7) of sample 'A', *A. niger, T. album and M.hiemalis* and of sample 'B' *Fusarium poae, Mucor hiemalis, T. album* and *Biosporomyces chlamydosporis* have been found.

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Enhanced Agar Oil Production through Application of Genetic improvement Tools

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ABSTRACT

Aquilaria species produces one of the important non-timber forest product agar oil, as a fungus infects its wood. However, fungal infection does not occur in all trees, as only few trees are susceptible to them. It happens due to genetic structure of the individual trees. Some trees resist, some infect from lesser to higher degree and some genuinely prone to infect from responsible fungus. The individual tree lie in the later type is quite useful and selected for this desirable traits. To make selection strict and fruitful, the index culling method (generally used for disease resistance) may be used for this purpose. Now, the selected trees multiplied vegetatively so that their desirable genetic structure can be copied as clones. These clones can be released in that locality for immediate production, simultaneously same clones should be put in clonal trial. The clonal trial will help in further selection of most desirable clones, which can be released for plantation.

Key-words : *Aquilaria*, Clones, Selection.

Role of *Zeuzera conferta* Walker in the initiation of agar lesions in *Aquilaria agallocha* Roxb.

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ABSTRACT

Harvested agar plants (*Aquilaria agallocha* Roxb.) from Kamrup, Golaghat and Jorhat districts were examined for determining the incidence of the stem borer, *Zeuzera conferta* Walker, and its association with agar lesion formation. Out of 43 samples examined in Golaghat and Jorhat districts, 34.88 % of the plants showed borer injuries and tunnels, while a lower incidence (15 %) of the insect was observed in plants from Kamrup district. Though agar lesion formation was recorded around majority of the insect tunnels in the wood, in 10.71 % of the samples from Golaghat and Jorhat, and 21.43 % of the samples from Kamrup district the tunnels were not associated with any agar lesion.

FEASIBILITY OF USING AGAR AS A SECONDARY HOST PLANT WITH SANDALWOOD UNDER RAINSHADOW BELT OF ASSAM

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ABSTRACT

To investigate the feasibility of agar as a secondary host plant with Sandalwood (Santalum album L) under the rainshadow belt of Assam, a long-term field experiment was undertaken at the experimental farm of the R&D centre, with five different species in the year 1997. Four years performance revealed that both plant height and collar diameter found appreciably higher with Vitex negundo, an indigenous wild medicinal plant followed by Aquilaria agallocha Roxb. a potential high value aromatic tree of this region. Considering the economic importance of the species, A. agallocha could be established as profitable secondary host plant for raising Santalum album L. plantation.

The initial growth performance indicates that in the foot hill area under rainshadow belt and piedmont upland both Sandal and Agar could be potential medicinal and aromatic plants in this region.

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FUNGI ASSOCIATED WITH THE DISEASED WOOD (AGARWOOD/ AGARU) Of *AQUILARIA AGALLOCHA* ROXB. (FAMILY THYMELEACEAE) GROWN IN ASSAM.

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ABSTRACT

Agarwood or Agaru of commerce is the diseased wood of *Aquilaria agallocha* Roxb. (Family-Thymelaeceae). Disease is caused by fungal infection. Infection of fungi in the wood of standing tree may occur due to insect bore by the larvae of *Zeuzera conferta* Walker. or due to mechanical injuries. The fungi *Botryodiploidia theobromae, Fusarium solani, Mucor hiemalis and Rosellina necatrix* have isolated from the diseased wood caused by insect boring. Over and above mentioned fungi, fungi like *Aspergillus granulatum, Penicillium citrinum* and a *Penicillium sp.* have also been isolated from the diseased wood caused by mechanical injuries.

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Investigation on the formation and development of agaru in *Aquilaria agallocha* Roxb.

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ABSTRACT

Deposition of oleoresin in *Aquilaria agallocha* Roxb. was studied in inner woody tissues after artificial inoculation with four different fungi viz., *Fusarium sp., F. oxysporum* (Isolate No. 1 and 2). *Trichoderma sp.* and one unidentified fungus isolated from infected Agar wood plants during the survey (fixed and roving) in and around Jorhat district of Assam. Plants above the age of seven years are only vulnerable for infection by the fungus responsible for oleoresin deposition. The association of the fungus (unidentified) was confirmed through Koch's postulates in 60% of the plants after inoculation. The fungus maintained its self-life (viability) more than a year in room temperature in both PDA and host extract medium.

Prospects of *Aquilaria agallocha* Roxb. Plantation in Arunachal Pradesh

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ABSTRACT

Arunachal Pradesh with its wide range of Forests and agroclimates offer necessary habitats for the growth of Agar. Till recently there were good population of this valuable tree in the state. But indiscriminate exploitation has drastically depleted its resources in the state. Realizing the eminent peril to the Agar tree, due to over exploitation and depletion of the forest the state Government has banned the exploitation of the agar wood from the forest. The forest department through its various schemes promotes cultivation of Non-timber Forest Produce including Agar. Such plantation can be taken up under community/ Joint Forest Management Schemes. The local institutes have the technology for agar plantation. The various Agroforestry systems and models proposed for the state to rehabilitate and improvement of the jhum lands and fallow can use Agar as one of the tree component. The integrated projects involving various departments of the state for the benefit of the local people can change the present scenario of scarcity of agar wood to plenty. The paper discusses the prospects of Agar Plantation in Arunachal Pradesh.

Some Observations on Agarwood Export Trade

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ABSTRACT

The sustainable and commercial use of Agarwood suffers from one single paradox. The secret of development of fungal patches in the Agar wood tree the nature has kept it to itself. So even if Agar wood trees are planted on a massive scale there is no guarantee that Agarwood can be harvested in commercial quantities.

Aquilaria malaccensis is included in Appendix II to the Convention on International Trade in Endangered Species of fauna and flora apart from included in the negative list of export in the current export policy. This implies that Agarwood and products obtained from the wild are prohibited to export.

The demand of Agarwood chips and dust in Arabian countries continue to rise. There had been instances in Mumbai port when Agarwood dust in huge quantities was attempted to be exported under a wrong or vague declaration. I am not sure whether these exporters can be called as a part of Agarwood trade as they were Mumbai based Merchant exporters dealing in many other products Agarwood being one of them.

As Agarwood trade is an export oriented the trade in its own interest should develop a method of distancing itself from such elements. A bad name is not a conductive situation for the development of the trade itself.

There is need for controlled use, apart from continuous monitoring and research on upgrading of database on silvicultural as well as trade aspects.

Scope & Dimension of Agar (*Aquilaria spp.*) plantation in North East region

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ABSTRACT

The scope of agar plantation in the N. E. Region and its socio-economic impact are discussed. The role of forest department in formulating the plantation techniques and provided them for private plantation may bring better results which may improve upon the economics of the region.

To augment agar oil and agaru formation a long term experiment has been laid out during 1980-81 rainy season with spacing 5m x 5m at Basistha, Guwahati. Among various cultural manipulations imposed upon as treatments on 24-01-86 insertion of nails inside the stem helped in the black formation around the nail but found only nearby it.

The author emphasized the fact that in absence of any clear-cut regulations on the trade practice much confusion arises and the traders many often being harassed. To avoid this there should have some clear-cut policy to guide the disposal and transit rules of Agar and Agar products.